

Eigenvalues and inverse problems

Where a symmetric stochastic matrix can be spectrally unique

Difficulty: challenging **Importance:** interesting to specialist**Status:** [Lean verified](#)

Rating rationale: Challenging reflects the geometry of isospectral stochastic families in arbitrary dimension; specialist impact concerns the narrow property of spectral uniqueness within that class.

Resolution — 2026-09-11

Negative resolution by Matthew J. Colbrook (Department of Applied Mathematics and Theoretical Physics, University of Cambridge). See the [complete manuscript](#), **Theorem IS-02, sections 1–2** ([PDF](#) · [LaTeX](#)), prepared 11 September 2026.

The order-four counterexample is real symmetric, nonnegative and stochastic, has spectrum $\{1, 1, 0, -1\}$ and positive trace, and is spectrally unique up to permutation. It lies outside every segment in the proposed locus, disproving the universal necessary condition at an allowed dimension.

The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently verified the full proof and its match to the exact target on 11 September 2026: [detailed PASS review](#). The review records a hash of the unchanged proof text. This is independent agent verification, not external human peer review or formal certification. [The submission history and diagnostic record](#) preserve the initial solution claim. The ratings above are historical, and the earlier literature checks below are retained.

Lean proof and verification evidence — 2026-09-14

The complete original necessary-locus conjecture has a Lean-verified negative answer. The complete spectral-uniqueness locus implication is disproved at order four. The actual symmetric stochastic witness has positive trace and the specified characteristic polynomial. Every admissible isospectral competitor is proved permutation-similar to it, while the witness is excluded from every proposed closed segment and from the actual extreme points. No competing matrix is restricted to a preselected support pattern.

Immutable formal proof. The checked exports are *NLA.IS02.witness_certificates*, *NLA.IS02.spectral_uniqueness*, *NLA.IS02.locus_exclusion*, *NLA.IS02.counterexample*. Lean 4.33.1 uses pinned Mathlib *0df444a3* and LeanCert *621a43d7*; [complete pins and reproduction instructions](#) are retained. All four exported transitive axiom closures contain only *propext*, *Classical.choice* and *Quot.sound*.

Formalization: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, with OpenAI Codex assistance. Matthew J. Colbrook retains mathematical authorship. Two independent statement reviews preceded implementation, and two independent final proof reviews passed. Kernel-only LeanCert and [fresh isolated Linux Comparator and kernel verification](#) passed, including rejection controls. [Original artifact, exact checked inputs and independent operational audit](#) are retained. This is AI-assisted formal verification, not external human peer review or source-author endorsement.

Problem statement

For $n \geq 4$, put $\mathcal{S}_n = \{A \in \mathbb{R}^{n \times n} : A = A^T, A \geq 0, A\mathbf{1} = \mathbf{1}\}$ and $C_n = (\mathbf{1}\mathbf{1}^T - I)/(n-1)$. A matrix $A \in \mathcal{S}_n$ is *spectrally unique* if every $B \in \mathcal{S}_n$ with the same eigenvalues, including multiplicities, satisfies $B = R^T A R$ for a permutation matrix R . Let $[X, Y] = \{(1-t)X + tY : 0 \leq t \leq 1\}$.

Prove or disprove the following necessary condition: every spectrally unique $A \in \mathcal{S}_n$ with $\text{tr } A > 0$ belongs to

$$[I, C_n] \cup \bigcup_{V \in \text{vert}(\mathcal{S}_n)} ([I, V] \cup [C_n, V]).$$

Here a vertex is an extreme point of the indicated convex polytope; it need not be a permutation matrix. Only the stated implication is asserted. This asks where recovering a nonnegative symmetric stochastic matrix from its spectrum can be unique up to relabeling, a different issue from mere spectral feasibility.

References

Mourad and Abbas, [2013 preprint](#), definitions in §1 and Conjecture 5.1, p. 10; [published article](#), *Linear and Multilinear Algebra* 63 (2015), 869–881.

Earlier status check — 2026-09-08

Searches for the exact title, *symmetric doubly stochastic Conjecture 5.1*, and author/title combinations with *counterexample* and 2026 found no resolution. The source solves order three, which is excluded from the remaining statement above.

Audit update — 2026-09-10

Rechecked [Mourad–Abbas](#), §5, Conjecture 5.1, against the trace and locus restrictions here. Searches for spectrally unique symmetric stochastic matrices and later work on that conjecture found no general answer. The order-three classification is outside the displayed unresolved dimensions, so it does not change this entry's status. Evidence remains historical rather than a recent explicit reaffirmation.